

PAVAN RAJ R

AI/ML Engineer • Computer Vision Specialist • NLP & Automation

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with industry experience at **Robert Bosch GmbH**, specializing in computer vision, industrial automation, and NLP systems. Skilled in building end-to-end ML pipelines using Python, OpenCV, TensorFlow, and PyTorch. Passionate about deploying scalable, production-ready AI solutions that reduce manual errors and improve operational efficiency.

WORK EXPERIENCE

AI/ML & Computer Vision Intern | Robert Bosch GmbH

Jan 2026 – May 2026 • Bangalore

- Engineered Glow Plug Inspection System using Masked Template Matching — automated QC on industrial lines, reducing manual inspection errors by ~85%
- Built real-time image processing pipelines (OpenCV + Pypylon SDK); implemented NMS reducing false positives by 30% in production environments
- Designed Tkinter GUI for live inspection, ROI selection & template management — deployed on factory floor for operator use
- Delivered timestamped CSV logging with PASS/FAIL traceability and multi-template 0°/180° rotation handling for consistent output

EDUCATION

- MCA — AI, ML & Data Science** 2024 – 2026 (Pursuing)
S-VYASA Deemed University, Bangalore
- BSc — Physics & Computer Science** 2022 – 2024 • CGPA: 7.88
Surana Autonomous College, Bangalore

KEY PROJECTS

Glow Plug Inspection System — Industrial Automation | Python, OpenCV, NumPy, Tkinter, Pypylon SDK

- End-to-end automated vision system: image capture → masked template matching → tray validation → PASS/FAIL logging
- ROI filtering + multi-orientation support improved detection precision by 40% over standard methods

AI Voice Assistant — NLP & Gemini API | Python, NLP, SpeechRecognition, PyQt5, Gemini API, Multithreading

- Real-time voice assistant with continuous listening, Google Gemini API for context-aware NLP responses, and system automation features
- Multithreaded PyQt5 UI with modular architecture separating GUI, audio, and AI logic for scalability

Real-Time Face Recognition & Emotion Classification | Python, OpenCV, DeepFace, LBPH, NumPy

- Live webcam system combining LBPH face recognition + DeepFace multi-class emotion classification (happy, sad, angry, neutral)
- Dynamic user registration without full dataset retraining; modular train/inference scripts for clean deployment

Power BI Sales Analytics Dashboard | Power BI, DAX, SQL, Excel, Data Modelling

- Built interactive multi-page Power BI dashboard tracking revenue, churn, and KPIs across regions and product categories
- DAX measures for YoY growth, rolling averages & drill-through reports — cut monthly reporting time by 60%

CERTIFICATIONS

- GenAI Powered Data Analytics — Tata Consultancy Services via Forage (May 2026) | EDA, AI-driven delinquency prediction, data storytelling
- Operations Industrial Engineer Job Simulation — Siemens via Forage (May 2026) | Time studies, layout optimisation

TECHNICAL SKILLS

Languages	Python, Java, OOP Principles
AI / ML	TensorFlow, PyTorch, Scikit-Learn, Deep Learning, CNN, LSTM, Classification, Regression, Feature Engineering
Computer Vision	OpenCV, DeepFace, LBPH, Masked Template Matching, NMS, Pypylon SDK
NLP & Speech	NLP Pipelines, SpeechRecognition, Google Gemini API, Context-Aware Generation
Data & BI	Power BI, DAX, NumPy, Pandas, Matplotlib, Seaborn, SQL, MySQL
Tools & QA	Git, GitHub, Jupyter, PyCharm, VS Code, Selenium WebDriver, REST APIs

CORE COMPETENCIES

Problem Solving • Critical Thinking • Teamwork & Collaboration • Adaptability • Attention to Detail • Communication